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Quantitative Aptitude

Formula – Guide

Number System

Number System

Divisibility rule

- **Divisibility for 2 :** A number is divisible by 2 if its unit digit is 0,2,4,6 or 8.
- **Divisibility for 3 :** A number is divisible by 3 if the sum of its digits is completely divisible by 3.
- **Divisibility for 4 :** A number is divisible by 4 if the number formed using its last two digits is completely divisible by 4.
- **Divisibility for 5 :** A number is divisible by 5 if its unit digit is 0 or 5.
- **Divisibility for 6 :** A number is divisible by 6 if the number is divisible by both 2 and 3.
- **Divisibility for 8 :** A number is divisible by 8 if the number formed using its last three digits is completely divisible by 8.
- **Divisibility for 9 :** A number is divisible by 9 if the sum of its digits is completely divisible by 9.
- **Divisibility for 10 :** A number is divisible by 10 if its unit digit is 0.
- **Divisibility for 11 :** A number is divisible by 11 if the difference between sum of digits at odd places and sum of digits at even places is either 0 or is divisible by 11.
- **Divisibility for 12 :** The number which is divisible by both 3 and 4 is also divisible by 12.
- **Common divisibility rule for 7, 11, and 13 :**

- ➔ Make pairs of 3 digits from RHS.
- ➔ Add alternate pairs & take difference.
- ➔ If the resultant difference is divisible by 7, 11, or 13, then the original number is also divisible by these numbers.

Remainder Theorem

When we divide a polynomial f(x) by (x−c) the remainder is f(c). Then,

f(x) = (x - c) q(x) + r

i.e., Dividend = (Divisor × Quotient) + Remainder

Factor Theorem:

- When f(c) = 0 then x−c is a factor of f(x)
- When x−c is a factor of f(x) then f(c)=0

Unit Digit

- If the last digit of any number is 0,1,5, 6 and whatever its power ,then the unit digit of this number will remain the same.
- If the last digit of any number is 4 and its power is odd then the unit digit of this number is 4 and its power is even then the unit digit of this number is 6.
- If the last digit of any number is 9 and its power is odd then the unit digit of this number is 9 and its power is even then the unit digit of this number is 1.
- If the last digit of any number is 5 and multiplied with an odd number then the unit digit of this number is 5 and multiplied with an even number then the unit digit of this number is 0.

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Factorial

- $n! = n \times (n-1) \times (n-2) \times \ldots \times 3 \times 2 \times 1$
- $0! = 1$
- $1! = 1$

Factor

If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \ldots p_k^{ek}$,

Where p_1, p_2, p_3 p_k are prime numbers and e1, e2 and ek are their respective powers.

- **Total Number of Factors :**

  $(e_1 + 1) \times (e_2 + 1) \times (e_3 + 1) \times \ldots (e_k + 1)$

- **Total Number of odd Factors :** Odd factors do not include the prime factor 2. Therefore, in the prime factorization, ignore any factors of 2.

  Then, Total number of odd factors is = $(e_1 + 1) \times (e_2 + 1) \times (e_3 + 1) \times \ldots (e_k + 1)$

- **Total Even Factors :**

  Total Even Factors = Total Factors – Total Odd Factors

  Total Number of even factors = $e_1 \times (e_2 + 1) \times (e_3 + 1) \times \ldots (e_k + 1)$

- **Total sum of factors :**

  $S(N) = (1 + p_1 + p_1^2 + \cdots + p_1^{e1})(1 + p_2 + p_2^2 + \cdots + p_2^{e2}) \ldots (1 + p_k + p_k^2 + \cdots + p_k^{ek})$

- **Total sum of odd factor :**

  $S_{odd}(N) = (1 + p_1 + p_1^2 + \cdots + p_1^{e1})(1 + p_2 + p_2^2 + \cdots + p_2^{e2}) \ldots (1 + p_k + p_k^2 + \cdots + p_k^{ek})$

- **Total sum of even factor :**

  Total Sum of Even Factors = Sum of factors - Sum of odd factors

  $S_{even}(N) = (2 + 2^2 + \cdots + 2^{e0})(1 + p_1 + p_1^2 + \cdots + p_1^{e1}) \ldots (1 + p_k + p_k^2 + \cdots + p_k^{ek})$

- **Sum of reciprocal of all factor :**

  Sum of reciprocals of all factors = $\frac{Total\ Sum\ of\ all\ factors}{Number} = \frac{S(N)}{N}$

- **Product of factor :**

  Product of all factors = $N^{T(N)/2}$

- **Total number of perfect square factors :**

  Total perfect square factors = $(e_1/2 + 1) \times (e_2/2 + 1) \times \cdots \times (e_k/2 + 1)$

- **Total number of perfect cube factors :**

  Total perfect cube factors = $(e_1/3 + 1) \times (e_2/3 + 1) \times \cdots \times (e_k/3 + 1)$

- **Proper factor :**

  Total number of proper factors = n - 2

LCM and HCF

The product of two numbers = (The two numbers' HCF) x (The two numbers' LCM)

A x B = H.C.F.(A, B) x L.C.M.(A, B)

Here, A and B are the two integers.

HCF and LCM of Fractions:

$$\text{HCF of Fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}$$

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